



NEUTRONICS FINDINGS NOTE

DSR, Areal Density, and Yield Unfold: OpenMC Transport vs NeSST on N210808

For: Kyle Keipper, Will**From:** Prof. Tom Mehlhorn (MECS)**Run:** 090425_KK_opt_2burn (N210808 clean-1D burn), bang-time snapshot, ENDF/B-VIII.0, 400k particles, neutron + photon transport**Date:** 2026-07-30, updated 2026-07-31 (added the self-consistent first-principles coefficient and the final pipeline DSR after the traversed-pR switch)

Bottom line

Three methods on the same Helios snapshot agree once the areal-density definition is pinned down. The OpenMC / NeSST “discrepancy” was the Method-1 vs Method-2 distinction from Kyle’s slide, now resolved and wired into the pipeline: **run_analysis STEP 5b now reports DSR 4.52% (fuel) / 5.02% (+carbon) on the emission-weighted (traversed) pR, matching OpenMC (4.3%).** A first-principles reproduction of the NIF yield unfold shows the standard 4-DSR exponent is design-specific (~5 for this HDC-ablator target).

1. Two areal densities

quantity	OpenMC (transport)	NeSST (as shipped)	analytic Method-2
ρR_D (g/cm ²)	0.372	0.567	0.297
DSR (10-12 / 13-15)	4.31%	10.07% (fuel 9.51%)	—

NeSST $\rho R_{\text{emission}} = \text{center-to-edge } \int_0^R \rho \, dr$ (Kyle Method 1). OpenMC and the analytic Method-2 (isotropic 4π angle-averaged chord integral, validated: central source $\rightarrow$ p-R, uniform $\rightarrow$ 0.75 $\times$) give the emission-weighted traversed pR. Both traversed values (0.30–0.37) are ~half the center-to-edge (0.57); DSR follows (4.3% vs 10%). Abu-Shawareb: “the areal density of the system is emission weighted to larger radii, which is why...the areal density decreases as the yield increases” — so DSR tracks the traversed pR (Method 2), not center-to-edge. Residual OpenMC (0.372) vs Method-2 (0.297) is multiple-scatter + energy-dependent σ (transport-only).

Pipeline run (STEP 5b, traversed mode) on the same snapshot: **traversed pR D = 0.298, T = 0.447, C = 0.664 g/cm²** (fuel D+T = 0.745, total = 1.409); DSR 4.52% fuel / 5.02% total. The pipeline traversed ρR_D (0.298) reproduces the analytic Method-2 (0.297) to 1%.

Action: NeSST is unchanged (a pure pR $\rightarrow$ DSR mapper). We changed our code: `_composition_scatter` now feeds it the traversed pR (new `target_composition.traversed_species_areal_densities`); center-to-edge is retained and selectable via `rhoR_mode`.

1a. First-principles DSR $\leftrightarrow$ pR coefficient (self-consistent)

The empirical NIF calibration is $\rho R = 20.4 \times \text{DSR}$ (g/cm² per unit DSR). Our first-principles value must use the **same areal-density convention on both sides**. Using the traversed (Method-2) pR that the DSR is now derived from:

coefficient	value (g/cm ² per DSR)	definition
total (D+T+C)	28.1	total traversed pR / total DSR (1.409 / 0.0502)
fuel (D+T)	16.5	fuel traversed pR / fuel DSR (0.745 / 0.0452)

These bracket the empirical 20.4, as expected for a clean over-igniting 1-D burn where pR/DSR run high.

Reporting fix: the pipeline previously printed a single “31.4” that mixed conventions — center-to-edge fuel pR (1.42) over the traversed fuel DSR (0.0452). That number was neither Method 1 nor Method 2 and has been corrected to the two self-consistent values above (neutron_scatter now uses the traversed convention on both sides). The empirical relation is tied to the total, emission-weighted pR, so 28.1 is the value to compare against 20.4; the fuel-only 16.5 is the internal D+T check.

2. First-principles yield unfold $Y_{\text{total}} = Y(13-15) e^{(4 \cdot \text{DSR})}$

quantity	value
escaping 13–15 MeV / born	0.786
$Y_{\text{total}} / Y(13-15)$, OpenMC (1/N)	1.272
paper $e^{(4 \cdot \text{DSR})}$	1.188
effective exponent k	5.6 (~5.1–5.2 after DD/TT correction)

$k \gg 4$ because $n-^{12}\text{C}$ elastic loses little energy per collision, pushing 14 MeV neutrons into 12–13 MeV (out of the unscattered window, above the 10–12 DSR window), which inflates scattered-out / DSR. Consequence: the standard 4·DSR unfold under-estimates total yield by ~7% (1.19 vs 1.27) for a heavy-ablator target. A DT-only source run would pin the exponent exactly.

3. Gamma signature

0.0070 photons / source neutron escaping, ~540:1 dominated by $^{12}\text{C}(n,n'\gamma)$ 4.4 MeV (6.8×10^{-3}) over H(n, γ) 2.2 MeV (1.3×10^{-5}). The 4.4 MeV line is a clean HDC-ablator carbon signature.

Caveats

Clean-1D burn ($Y \sim 15.6$ MJ vs 1.37 measured); pR/DSR run high vs the real shot (published DSR 2.87%). This demonstrates the method + corrections, not an experimental match. One snapshot. Surface-tracking warnings are negligible (DSR stable to ~3% across two geometry rebuilds). $\sigma_{\text{el,D}} = 0.641$ b from ENDF/B-VIII.0 at 14.06 MeV.

Recommendations

1. Use the emission-weighted (Method-2 / OpenMC) pR against the measured DSR (pipeline now default).
2. Don't assume a 4·DSR unfold for carbon ablators; derive it per design (~5 here).
3. Pin which pR the experimental DSR $\leftrightarrow$ pR calibration is tied to (the emission-weighted total, per the 28.1 vs 20.4 comparison above).
4. Next: a degraded / under-igniting N210808 (lower pR) to map how the Method-1/2 gap, the unfold exponent, and the DSR $\leftrightarrow$ pR coefficient scale with yield.

Reproduce: examples/dump_snapshot.py (Helios env) $\rightarrow$ examples/compare_openmc_nesst.py (openmc-env).